

TECHNICAL SPECIFICATIONS

Medical Supplies Ukraine

HYSTEROSCOPIC COMPLEX (FULL RACK) WITH BIPOLAR RESECTOSCOPE 27 FR (9 MM)

1. General system requirements

- Equipment manufacturer - European countries or the USA.
- Condition: The equipment is new, unused, and manufactured no earlier than 2025.
- Platform: The video system, LED light source, and fluid delivery pump must be from a single hardware line from a single manufacturer to ensure full digital integration, display of pressure and flow parameters on the monitor, and synchronized control.
- Sterilization: All parts of the instruments that come into contact with the patient must withstand repeated sterilization in an autoclave at a temperature of 134°C.

2. Visualization and monitoring

2.1. 4K UHD video system (1 piece).

- Video processor and camera resolution: at least 3840×2160 pixels.
- Camera head: CMOS sensor, equipped with at least 3 programmable buttons for remote control.
- Chamber design: completely waterproof, autoclavable at 134°C.

2.2. Medical monitor (1 piece).

- Diagonal: at least 31 inches.
- Resolution: 4K UHD (3840×2160 pixels).
- Viewing angle: at least 178° vertically and horizontally.

2.3. Light source (1 piece).

- Type: LED.
- Power: equivalent to a xenon light source with a power of at least 300 W.
- Automatic lighting brightness adjustment function.

2.4. Fiber optic light guide (1 piece)

- Diameter: from 4.5 mm to 5.0 mm.
- Length: at least 250 cm.

3. Electrosurgical unit and pump

3.1. Electrosurgical generator (1 piece).

- Maximum output power: at least 300 watts.
- Availability of a specialized mode for bipolar hysterectomy in saline (0.9% NaCl).
- Constant monitoring of work safety.

3.2. Pedal for electrosurgical generator (1 piece).

- Two-button pedal for activating cutting and coagulation modes.

3.3. Hysteropump / endopump (1 piece).

- Functionality: automatic fluid supply and aspiration.
- Operating pressure support: in the range from 30 to 150 mm Hg.
- Flow rate: up to 500 ml/min.
- Integrated fluid deficit monitoring system (difference between supplied and pumped fluid).

4. Bipolar hysteroscopes 27 Fr (9 mm) (1 assembly). Consists of the following mandatory components:

- Rigid optics: diameter 4.0 mm, viewing angle 12 degrees (1 piece).
- Outer sheath 27 Fr (9.0 mm) with perforation for continuous fluid outflow (1 piece).
- Inner tube with ceramic beak insulation (1 piece).
- Standard obturator for safe insertion (1 piece).
- Passive working element (1 piece).

4.1. Tools and cables

- Bipolar high-frequency cable for connecting the working element to the generator (1 piece).
- Reusable 2-liter container for collecting aspiration fluid (2 pieces).

5. Complete set of bipolar electrodes

All electrodes must be compatible with a 27 Fr working element and designed to work in saline (0.9% NaCl).

- Standard bipolar loop electrode at an angle of 90° - for cutting layers of tissue (10 pieces).
- Bipolar loop electrode at an angle of 45° - for specific angular resections (5 pieces).
- Straight bipolar loop electrode at an angle of 0° - for frontal cutting (2 pieces).
- Bipolar ball electrode - for vascular coagulation (2 pieces).
- Bipolar knife electrode (needle/tip) - for precise dissection of synechiae or septa (2 pieces).
- Bipolar vaporization electrode (button/mushroom) - for plasma vaporization (2 pieces).
- Bipolar roller electrode (barrel) - for endometrial ablation (2 pieces).

6. Intrauterine morcellation system / shaver (1 set)

6.1. Hardware part of the shaver

- Motor control unit (1 piece): digital console for controlling speed and direction of rotation (constant rotation and oscillation).
- Shaver motor handle / drive (1 piece): ergonomic, fully sealed, with brushless motor and autoclavable at 134°C.
- Shaver control pedal (1 piece): to activate rotation and switch modes.

6.2. Shaver tool part

- Rigid optics for morcellation (1 piece): specialized optics adapted to work with the morcellator tube to provide direct visual control of the blade operation (angle of view from 0° to 12°).
- Operating tube for morcellation (1 assembly): set (outer and inner tubes) with a wide straight working channel for inserting a shaver blade and simultaneous aspiration (1 piece), obturator for safe insertion (1 piece).
- Disposable cutting attachments / shaver blades (10 pieces): sterile morcellator blades with an internal channel (window) for simultaneous cutting and active aspiration of dense tissues (fibroids, polyps).

7. Disposable materials and infrastructure

7.1. Disposable consumables

- Sterile systems (tubes) for fluid delivery (50 pieces).
- Sterile aspiration tubes (50 pieces).

7.2. Medical rack / trolley (1 piece)

- Hardware design: metal, at least 4 shelves for placing devices.
- Antistatic wheels with braking mechanisms.
- Built-in cable management system (hidden wiring).
- Bracket for secure mounting of a medical monitor.

8. Other requirements and related services

- Warranty service: the warranty period for all proposed equipment must be at least 12 months from the date of commissioning.
- Installation and training services: The vendor should deliver, assemble, install, and commission the equipment at the workplace, as well as instruct / train the medical and technical personnel of the hospital on the rules for operating the equipment.

SPECIFICATIONS OF LAPAROSCOPIC SYSTEM

1. General requirements

- Equipment manufacturer - European countries or the USA.
- The equipment is new, has not been used.
- Year of manufacture no earlier than 2023.
- All components of the same platform or fully integrated with each other.
- Unified system control interface.
- Availability of a centralized digital control bus for compatible laparoscopic platform modules, with the ability to control connected devices from the camera, the main unit or through a central communication module.

2. Video system 4K NIR/ICG

- True 4K UHD resolution (without upscaling).
- NIR/ICG fluorescence with real-time overlay.
- Display modes: white light / combined / monochrome.
- Signal delay no more than 50 ms.
- Digital image enhancement algorithms.
- At least 2 independent 4K video outputs.

3. Camera

- At least 3 CMOS sensors or equivalent.
- Resolution 3840×2160.
- Integrated NIR/ICG technology.
- Programmable buttons on the camera head - at least 3.
- Autoclavable up to 134°C.
- Ability to control compatible platform components from camera head buttons via a central communication interface.

4. Source of light

- LED with a power of at least 300 W or equivalent.
- NIR spectrum support.
- Stable light flux without pulsation.
- Automatic intensity adjustment.
- Compatibility with a centralized video endoscopy platform control system.

5. Fiber optic (light cable)

- Diameter 4.5-5 mm.
- Length not less than 300 cm.
- High light conductivity.
- Minimal loss of light flux.
- Resistance to repeated sterilization.
- Compatibility with light source and endoscope.

6. Endoscopic optics

- Telescope 10 mm - 1 piece.
- Viewing angle 30° - 1 piece.
- Compatible with 4K and NIR/ICG.

- Anti-reflective coating.
- High light transmittance.
- Accurate color reproduction.

7. Medical trolley (stand)

- At least 4 shelves.
- Antistatic wheels with locking.
- Cable management system.
- CO₂ cylinder holder.
- Bracket for medical monitor.
- The design of the trolley must provide for the placement of integrated video system modules and the connection of a centralized control bus.

8. Medical monitor

- Diagonal size not less than 31".
- 4K UHD.
- Brightness of at least 800 cd/m².
- Contrast ratio of at least 1000:1.
- Viewing angle of at least 178°.
- Response time no more than 10 ms.

9. CO₂ insufflator

- Gas flow is at least 40 l/min.
- Pressure range 1-30 mmHg.
- Automatic maintenance of the set pressure.
- Gas heating.
- CO₂ filtration.
- Overpressure protection.
- Compatible with reusable and disposable trocars.
- Possibility of integration into a centralized endoscopic platform control system and display of parameters on the central video unit / communication module.

10. Coagulator

- Power not less than 300 Watts.
- Monopolar and bipolar modes.

- Neutral electrode control.
- Control pedal.
- Compatible with laparoscopic monopolar and bipolar instruments.

11. Aspiration and irrigation system

- Combined suction and irrigation system.
- Suction capacity of at least 30 l/min.
- Irrigation function (fluid supply).
- Vacuum and liquid supply adjustment.
- Availability of a control pedal.
- Containers for collecting liquid.
- Ability to work as part of an integrated endoscopic platform.

12. Reusable laparoscopic instruments

- Laparoscopic graspers 5 mm, working length not less than 330 mm, atraumatic, with collapsible design - 4 pieces.
- Maryland type laparoscopic dissectors, 5 mm, working length not less than 330 mm, collapsible - 2 pieces.
- Laparoscopic scissors, straight and/or curved, 5 mm, insulated, collapsible - 2 pieces.
- Laparoscopic needle holders, 5 mm, working length not less than 330 mm, with a retainer - 2 pieces.
- Laparoscopic bipolar forceps, straight and curved, 5 mm, collapsible - 2 pieces.
- Monopolar laparoscopic hook, 5 mm - 1 piece.
- Monopolar laparoscopic blade, 5 mm - 1 piece.
- Clip applicator for polymer clips, reusable, laparoscopic - 1 piece.
- Clip applicator for titanium clips, reusable, laparoscopic - 1 piece.

13. Cables and electrodes

- Monopolar cable with a length of at least 3 m - 1 piece.
- Bipolar cable - 1 piece.
- Neutral electrode cable - 1 piece.
- Compatible with electrosurgical generator and laparoscopic instruments.

14. Consumables

- Polymer clips - at least 50 pieces.

- Titanium clips - at least 50 pieces.
- Disposable trocars 5 / 10 / 12 mm - set.
- CO₂ insufflation tubes - set.
- Containers for the aspiration and irrigation system - set.
- Irrigation systems (hoses, tips, connecting lines) - set.
- Sterile containers / lines for liquid supply - set.

15. Platform integration requirements

- All main modules of the system must work together as part of a single laparoscopic platform.
- Control of compatible devices should be possible through a centralized communication interface, including from camera or head unit controls.

16. Other requirements and related services

- Warranty service: The warranty period for all proposed equipment must be at least 12 months from the date of commissioning.
- Installation and training services: The vendor should deliver, assemble, install, and commission the equipment at the workplace, as well as instruct / train the medical and technical personnel of the hospital on the rules for operating the equipment.

SPECIFICATIONS OF ENERGY PLATFORM FOR VESSEL SEALING

1. General requirements

Equipment manufacturer - European countries or the USA.

The equipment is new, has not been used.

Year of manufacture - no earlier than 2023.

A single energy platform with automatic control of modes.

Support for bipolar vascular sealing technology with tissue impedance control.

2. Energy platform

Isolated output.

Automatic adaptation of energy to the tissue.

Operating frequency - about 400 kHz.

Maximum power not less than 250–270 W.

Automatic completion of coagulation cycle.

Audible and visual indication of completion.

3. Integration and control

Support for centralized control (SCB or equivalent).

Control via foot pedal or instrument handle.

Automatic recognition of instruments (RFID or equivalent).

LED system status indication.

4. Laparoscopic instruments

Universal instrument (5/10 mm, ≥ 35 cm).

Sealing of vessels up to 7 mm.

Non-stick branches.

360° rotation.

Coagulation and dissection function.

Maryland type instrument:

Diameter 5 mm, length ≥ 35 cm.

Precise dissection and coagulation.

Atraumatic instrument:

For gripping tissues.

Atraumatic branches.

5. Open instruments

Bipolar forceps for open surgery.

Sealing of vessels up to 7 mm.

Platform compatibility without adapters.

6. Types of instruments

Availability of disposable sterile instruments

or reusable instruments with a service life of at least 20–50 cycles.

Sterilization capability (ETO or low temperature).

7. Safety

Leakage current monitoring according to IEC 60601.

Overheating protection.

Audio and LED error signals/tone.

Blocking with incompatible instruments.

8. Technical parameters

Power supply: 90–264 V, 48–62 Hz.

Maximum current: up to 5 A.

Operating temperature: 10–40°C.

Humidity: 30–75%.

Dimensions: about 300×377×105 mm.

Weight: about 5 kg.

9. Other requirements and related services

- Warranty service: The warranty period for all proposed equipment must be at least 12 months from the date of commissioning.

- Installation and training services: The vendor should deliver, assemble, install, and commission the equipment at the workplace, as well as instruct / train the medical and technical personnel of the hospital on the rules for operating the equipment.